Heaven's Light is Our Guide

Rajshahi University of Engineering and Technology

Department of Mechatronics Engineering

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Name of Experiment	:	Study of Zener Diode
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Experiment No: 01

Name of Experiment: Study of Zener Diode

1.1 Objectives

1. To study the forward and reverse bias characteristics of a Zener diode.
2. To determine the Zener voltage V_Z and observe the breakdown region.
3. To measure the current I_Z through the diode for different applied voltages.
4. To plot the V_Z-I_Z characteristic curve and verify the voltage regulation property of the Zener diode.

1.2 Theory

A Zener diode is a specially doped silicon diode designed to operate in reverse breakdown without being damaged. Unlike an ordinary diode, which is mainly used in forward bias, the Zener diode is intentionally used in the reverse-bias region where the voltage reaches a specific value known as the Zener voltage, V_Z. At this point, the current rises sharply while the voltage across the diode remains nearly constant. This property makes it useful for voltage regulation and reference applications [1], [2].

In forward bias, a Zener diode behaves like a normal PN junction diode and begins to conduct once the applied voltage exceeds approximately $0.7\,\mathrm{V}$ for silicon. In reverse bias, it allows only a small leakage current until the breakdown region is reached. Once the reverse voltage crosses the knee of the characteristic curve, the current increases rapidly without a significant change in the diode voltage. This nearly vertical region is the basis of voltage regulation, because the output voltage remains close to V_Z even when the current varies over a wide range [1], [3].

The Zener diode characteristic has three main regions: forward conduction, reverse leakage, and reverse breakdown. The complete I-V relationship is shown in Figure 1.2. In the breakdown region, the voltage is almost constant, so the diode can be used as a stable reference source. That is why Zener diodes are commonly used in regulator circuits, clipping circuits, and protection networks [2].

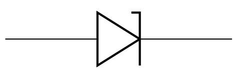


Figure 1.1: Zener diode symbol.

Figure 1.1 shows the standard symbol of a Zener diode. It is similar to a normal diode symbol, but the cathode side is bent to indicate its reverse-breakdown operation. The actual device behavior is commonly explained using its I-V characteristic, which helps to identify the forward region, the leakage region, and the breakdown region.

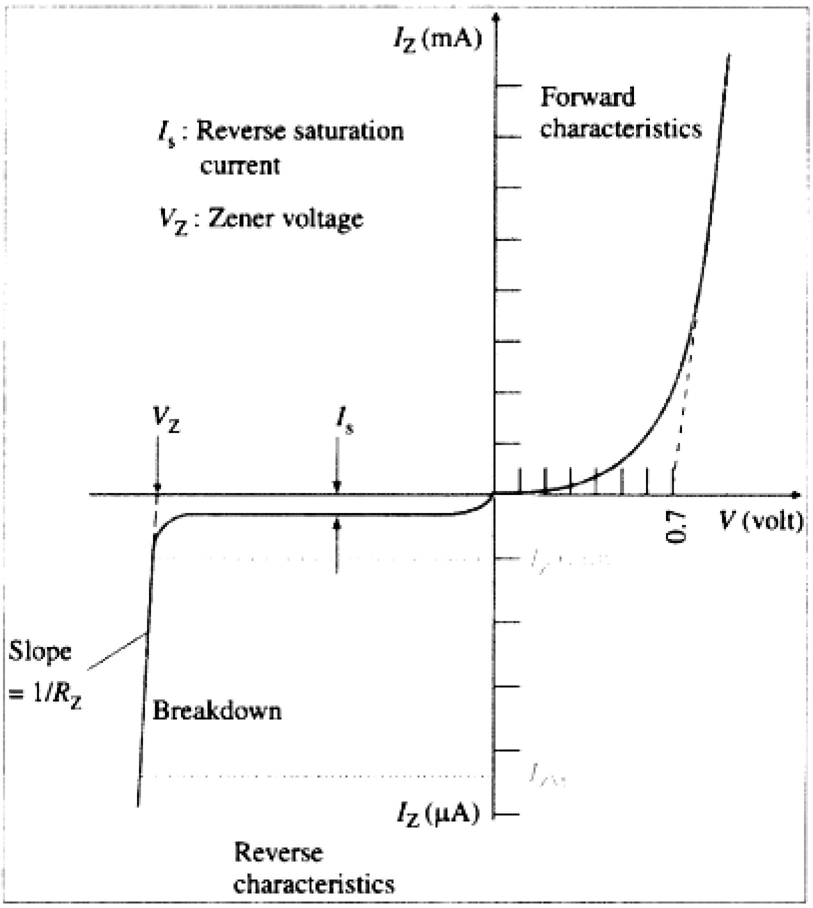


Figure 1.2: I-V characteristic curve of a Zener diode [4].

A useful approximation for analysis is to treat the Zener diode as an ideal voltage source of value V_Z. This is called the first approximation, as shown in Figure 1.3. In this model, the voltage across the diode is assumed to remain constant whenever it is in the breakdown region, regardless of the current through it.

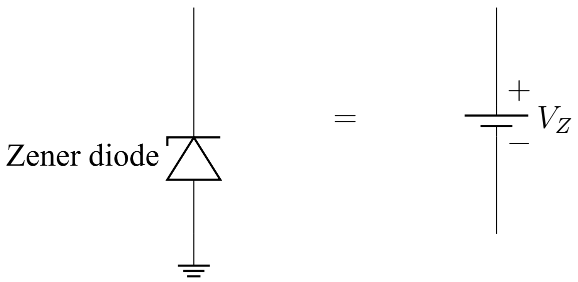


Figure 1.3: First-approximation model of a Zener diode.

A more realistic model is the second approximation, where the diode is represented as an ideal voltage source V_Z in series with a small dynamic resistance R_Z, as shown in Figure

1.4. This additional resistance accounts for the slight change in voltage that occurs as current increases within the breakdown region. Since R_Z is usually small, the output voltage remains close to the nominal Zener value and the device continues to behave as a practical voltage regulator [3].

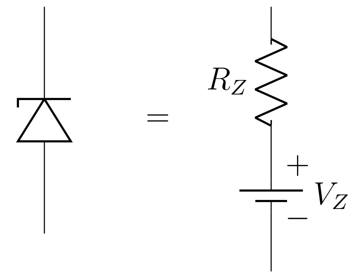


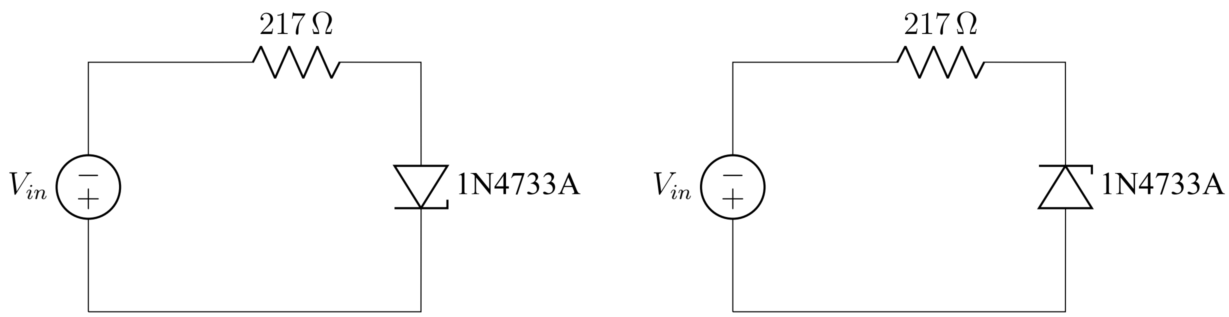
Figure 1.4: Second-approximation model of a Zener diode.

This experiment is designed to study the forward and reverse characteristics of a Zener diode and to verify how it maintains a nearly constant voltage under varying current conditions.

1.3 Apparatus Required

1. Zener Diode (Model: 1N4733A)
2. DC Regulated Power Supply
3. Resistor: $217\,\Omega$
4. Multimeter
5. Breadboard
6. Jumpers

1.4 Circuit Diagram



(a) Circuit Diagram of Forward Bias

(b) Circuit Diagram of Reverse Bias

1.5 Data Table

Here, $R_1 = 217\,\Omega$

V (Volt)	V_R (Volt)	V_Z (Volt)	$I_Z = \frac{V_R}{R_1}$ (mA)
0.1	0	0.09	0
0.3	0	0.29	0
0.5	0	0.49	0
0.7	0.01	0.68	0.06
0.9	0.13	0.75	0.60
1.1	0.30	0.77	1.38
1.3	0.46	0.79	2.15
1.5	0.64	0.79	2.95

Table 1.1: Data for Forward Bias

V (Volt)	V_R (Volt)	V_Z (Volt)	$I_Z = \frac{V_R}{R_1}$ (mA)
1	0	0.99	0
2	0	1.99	0
3	0	3.00	0
4	0	3.99	0
5	0.08	4.94	0.34
6	0.66	5.33	3.03
7	1.62	5.36	7.46
8	2.40	5.38	11.06
10	4.56	5.41	21.01

Table 1.2: Data for Reverse Bias

1.6 Calculation

Given,

$$R_1 = 217\,\Omega$$

The zener current is calculated using

$$I_Z = \frac{V_R}{R_1} \tag{1.1}$$

For Forward Bias

$$I_Z = \frac{0.01}{217} \times 10^3 = 0.046\,\text{mA} \approx 0.06\,\text{mA}$$

$$I_Z = \frac{0.13}{217} \times 10^3 = 0.600\,\text{mA} = \boxed{0.60\,\text{mA}}$$

$$I_Z = \frac{0.30}{217} \times 10^3 = 1.382\,\text{mA} = \boxed{1.38\,\text{mA}}$$

$$I_Z = \frac{0.46}{217} \times 10^3 = 2.119\,\text{mA} \approx 2.15\,\text{mA}$$

$$I_Z = \frac{0.64}{217} \times 10^3 = 2.949\,\text{mA} = \boxed{2.95\,\text{mA}}$$

For Reverse Bias

$$I_Z = \frac{0.08}{217} \times 10^3 = 0.369\,\text{mA} \approx 0.34\,\text{mA}$$

$$I_Z = \frac{0.66}{217} \times 10^3 = 3.041\,\text{mA} = \boxed{3.03\,\text{mA}}$$

$$I_Z = \frac{1.62}{217} \times 10^3 = 7.465\,\text{mA} = \boxed{7.46\,\text{mA}}$$

$$I_Z = \frac{2.40}{217} \times 10^3 = 11.06\,\text{mA} = \boxed{11.06\,\text{mA}}$$

$$I_Z = \frac{4.56}{217} \times 10^3 = 21.01\,\text{mA} = \boxed{21.01\,\text{mA}}$$

1.7 Graph

The Zener diode characteristic has been plotted separately for the forward and reverse bias regions because the two regions operate over very different voltage and current ranges. This allows the knee of the forward characteristic and the breakdown region of the reverse characteristic to be observed more clearly.

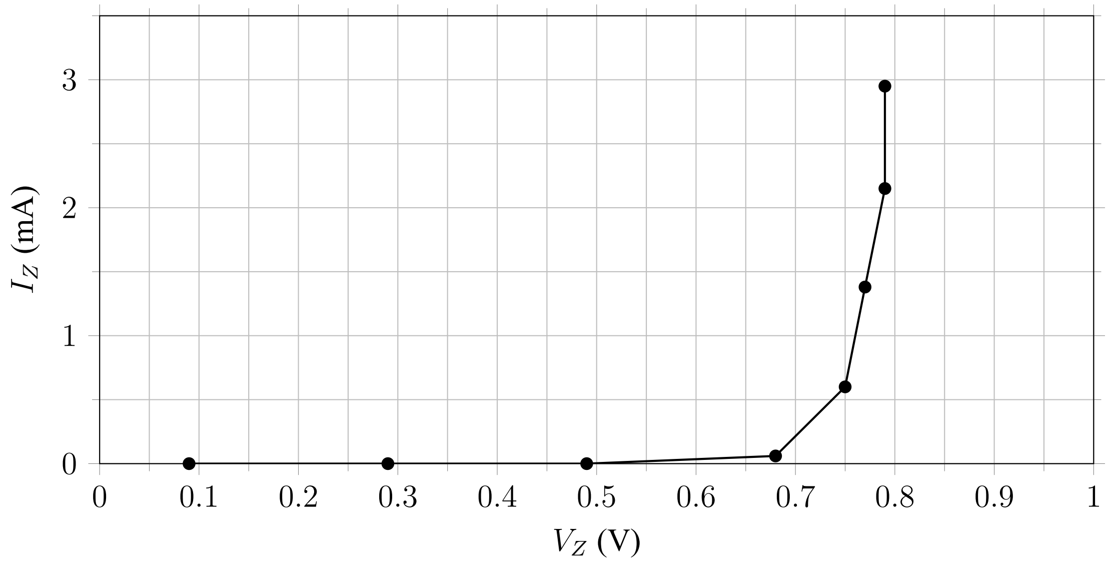


Figure 1.6: Forward-bias characteristic curve of the Zener diode.

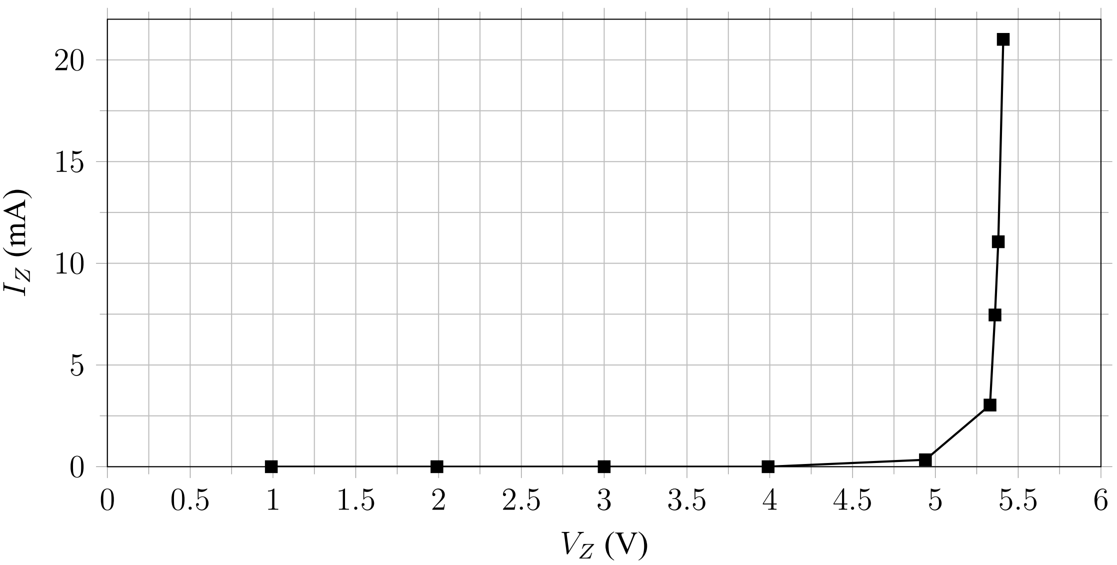
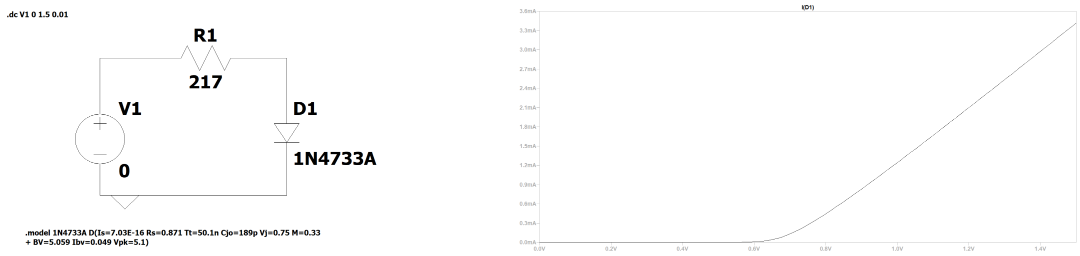


Figure 1.7: Reverse-bias characteristic curve of the Zener diode in breakdown region.

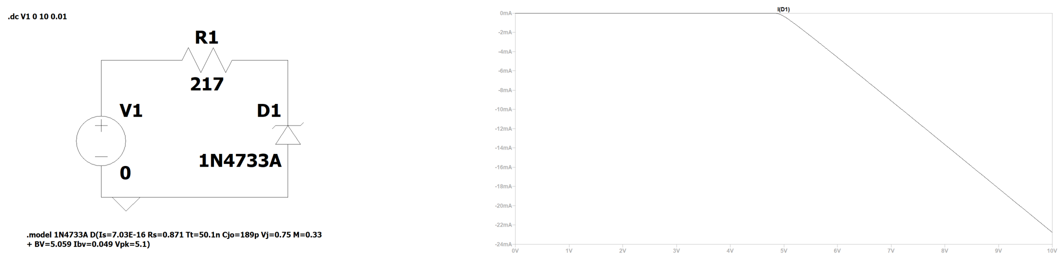
From the two graphs, it is observed that the forward current rises sharply after a small forward voltage, while in reverse bias the voltage stays nearly constant at about $5.4\,\mathrm{V}$ once the breakdown region is reached. This behavior confirms the Zener diode's use as a voltage regulator.

1.8 Simulaiton



(a) Schematic **(b)** Output

Figure 1.8: Simulation of Forward Bias: (a) Schematic and (b) Output.



(a) Schematic **(b)** Output

Figure 1.9: Simulation of Reverse Bias: (a) Schematic and (b) Output.

1.9 Discussion

The experimental data shows that a Zener diode behaves like an ordinary diode in forward bias, where the current remains very small at low voltages and then rises rapidly once the forward threshold is reached. In reverse bias, the current remains almost zero until the breakdown region begins, after which the voltage across the diode remains nearly constant while the current increases significantly. From the measured values, the diode voltage stays close to about $5.4\,\mathrm{V}$ in the breakdown region, which confirms the voltage regulation property of the Zener diode. The separate graphs for forward and reverse bias also show that the diode is suitable for maintaining a stable voltage under varying current conditions, although the resistance of the circuit and the applied supply must be chosen carefully to keep the diode within its safe operating range.

1.10 Conclusion

The experiment successfully demonstrates the characteristic behavior of a Zener diode in both forward and reverse bias. In forward bias, the current increases after a small threshold voltage, while in reverse bias the diode remains nearly non-conducting until breakdown occurs, after which the voltage remains almost constant at the Zener voltage. The measured results and the plotted characteristic curves confirm that the Zener diode can be used as a voltage regulator because it maintains a nearly constant output voltage even when the current changes over a wide range. Therefore, the objective of studying the Zener diode and observing its breakdown behavior was achieved.

References

[1] R. L. Boylestad and L. Nashelsky, *Electronic Devices and Circuit Theory*, 11th. Pearson, 2013.

[2] V. K. Mehta and R. Mehta, *Principles of Electronics*, 11th. S. Chand Publishing, 2012.

[3] Department of Electrical and Electronic Engineering, *Laboratory manual for the study of zener diode*, EEE 2188 Laboratory Manual, 2026.

[4] Sarthaks eConnect, *Explain the forward and reverse characteristic of a zener diode*, https://www.sarthaks.com/1392374/explain-the-forward-and-the-reverse-characteristic-of-a-zener-diode, 2026.